



Special Issue Editorial: Adaptive management for biodiversity conservation in an uncertain world

Uncertainty and adaptive management for biodiversity conservation

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ABSTRACT

Adaptive management of natural resources is widely supported, but in biodiversity conservation there have been few practical applications of the approach in its entirety. Some of the contributions to this special publication examine progress in the implementation of adaptive approaches into conservation policy, while others explore novel theoretical and modeling approaches that seek to accommodate the complexities of real-world applications. Several of the papers address the treatment of uncertainty in adaptive management through innovative approaches to experimentation and monitoring, use and characterisation of expert knowledge and reconciliation of differences of opinion about parameters or systems. Drawing on these contributions, we discuss the major impediments to implementing adaptive management, why adaptive management has been slow to be implemented and how this can be redressed.

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1. Introduction

Since Walter and Hilborn's (1976) and Holling's (1978) seminal work, the idea of adaptive management has gained worldwide interest and support as an approach to ecosystem management for conservation of biodiversity. Its key elements include explicit definition of management goals, development of plausible alternative management strategies to achieve those goals, implementation of two or more strategies in a comparative experimental framework to spread risks of management failure and improve understanding of system responses to management, monitoring to evaluate the relative merits and limitations of alternate strategies, and iterative modification of management strategies to improve management outcomes (Lindenmayer and Burgman, 2005). An important quality of the adaptive approach is that it deals with uncertainty through a structured improvement of relevant knowledge, while seeking to minimise risks associated with ongoing management, which inevitably arise from imperfect information about system response.

Despite the much-lauded strengths of this approach in both scientific literature and land management discourse, there are very

few examples where the approach has been applied in its entirety to real-world conservation problems. For example, many applications examine only one management option at a time, monitoring outcomes and changing strategy only when the favoured option has failed, an approach labelled 'trial and error management' (Duncan and Wintle, 2008). In spite of this, adaptive management has become central in the dialogue of natural resource management policy of most environmental agencies. Given this continued interest in applying adaptive management, a special issue bringing together recent developments in adaptive management science and its application in policy is timely.

Papers in this issue arose out of symposia at the International Ecology Conference held in Brisbane Australia, in August 2009. Contributions address both theoretical and practical aspects of adaptive approaches to dealing with uncertainty in ecosystem management for conservation. The subject matter spans a wide range of important conservation problems, landscape types and cultural settings. The contributions address terrestrial and freshwater applications that deal with management of invasive species, fire and water regimes, herbivores and predators, human exploitation of natural resources and threatened species management. The contributed papers include novel theoretical and modeling perspectives designed to accommodate the complexities of real-world applications of adaptive management. Several papers examine the treatment of uncertainty in adaptive management through

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innovative approaches to monitoring and experimentation, use and characterisation of expert knowledge, and reconciliation of divergent opinions about parameters or systems. Drawing on these contributions, we discuss the major issues in adaptive management, why adaptive management has been slow to be implemented and finally how impediments to adaptive management can be overcome.

2. Major issues in adaptive management

A wide variety of management approaches are now called adaptive management, ranging from any management in which actions or approaches change through time (by whatever means), to a more traditional interpretation that demands integration of models, active experimentation, monitoring, decision analysis and a mix of alternative management actions, whose progressive development is informed through learning by doing. Examples of real-world applications that incorporate all of the latter elements are still scarce. Case studies and reviews that address applications of the approach have revealed many problems that can cause failure of adaptive management programs (e.g. Walters 1997, 2007; Stringer et al., 2006; Duncan and Wintle, 2008; Allen and Stankey, 2009). Contributions in this issue address many of those problems in innovative ways. Some take an overview of practical attempts to apply adaptive management in real-world contexts, while others focus on specific aspects of the adaptive management framework and present new analytical approaches to address them, often with an illustrative example. This section provides an overview of the most troubling problems, and identifies which papers address each of them in Table 1.

2.1. Embracing uncertainty

Successful adaptive management requires willingness to “embrace uncertainty” rather than pretending that policy consequences are largely known so that only minor corrective action might be needed. This means there needs to be a policy commitment, most often driven by quantitative model development, for identifying alternative hypotheses about responses to management policies and for moving from passive to active (experimental) adaptive policy (Briceño-Linares et al., 2011; Kingsford et al., 2011; van Wilgen and Biggs, 2011). Runge et al. (2011) develop the concept of ‘expected value of information’ to help identify and resolve uncertainties that most impede the achievement of management objectives. Conroy et al. (2011) highlight the need to embrace uncertainty in biotic responses to climate change by implementing strategies that incorporate and test multiple plausible hypotheses,

rather than allowing the need for discovery to forestall management decisions.

2.2. Imaginative synthesis

We really do not know how to model complex ecological systems and to predict their vulnerabilities to various human policy interventions. There is a critical need to develop both innovative modeling approaches that aid in recognition of critical uncertainties and hypothesis development, and for identification of novel and win–win policy options. Briceño-Linares et al. (2011) recognise the importance of interacting social and ecological systems in their model of an endangered parrot population. Consequently, their management strategies involve an imaginative mix of educational programs, social initiatives and targeted employment opportunities with traditional habitat management. Both Howes et al. (2010) and Rumpff et al. (2011) explored the use of Bayesian networks to synthesise formal experimental data with informal knowledge within process models that describe the uncertain responses of woodland systems to both environmental variability and alternative management actions. In both studies, the authors elicited information from managers to construct initial process models, then updated them using new knowledge obtained by monitoring the outcomes of alternative management treatments at sites in a range of different initial states. While, Howes et al. (2010) were able to increase confidence in the predicted responses of their system, Rumpff et al. (2011) found that their updated models produced transition probabilities that differed markedly from those of initial models based on expert opinion, and hence demonstrated a substantial learning gain during the AM process. Where the modeling phase of adaptive management policy development leads to consistently poor or highly uncertain predicted performance for existing or baseline policy choices, the resulting sense of desperation among adaptive management decision makers can stimulate a generation of novel hypotheses and the discovery of imaginative policy options.

2.3. Experimental design

Most adaptive management programs end up having at least some experimental component aimed at “probing for opportunity” to improve management performance using policy choices as experimental treatments. It is misleading in general to pretend that simply monitoring responses to “best practice” policies over time will give definitive answers about how to improve those policies, since the policy responses are typically confounded with responses to other, uncontrolled factors that drive ecological change. An opportunistic mix of experiments, supplemented by expert

Table 1

Adaptive management issues addressed by papers in this issue. See text for description of the issues represented by table columns.

Paper	Embracing uncertainty	Imaginative synthesis	Experimental design ^a	Institutional barriers	Modeling for screening	Innovative monitoring	Conflict resolution ^b	Self-serving behaviour
Briceño-Linares et al.	X	x	T			x	E	
Colyvan et al.				x	x		G	
Conroy et al.	X	x	O		x	x	M	x
Kingsford et al.		x		x			H	
Probert et al.	X				x		M	
Rumpff et al.	X	x			X			
Runge et al.	X	x	F,O		x		M,V	
van Wilgen and Biggs	X	x		x		x	N	x

X – major focus, x – minor focus.

^a Experimental design type: T – titrate with progressively more severe treatments; F – factorial or pair-wise comparison of treatments; O – opportunistic mixture of experiments and spatial/temporal comparisons.

^b Approach to tradeoffs and conflict resolution: G – game-theoretic approach to identifying cooperative solutions; M – multi-attribute utility function; E – economic incentives for cooperation; H – hierarchies of objectives; V – value of information assessment (expected value of perfect information, EVPI); N – ongoing negotiation among stakeholders.

knowledge, is often applied to evaluate the assumptions underlying alternative strategies (Rumpff et al. (2011)). Even simple experimental designs, like paired comparisons of alternative treatments, can go a long way toward separating policy effects from other causes of ecological change (Mackenzie and Keith, 2009). A landscape experiment in Kruger National Park, for example, was established to assess tree responses in patches maintained with different densities of elephants, advancing on previous *laissez faire* culling policies that failed to deliver improvements in knowledge about critical elephant densities associated with decline in tree canopy cover (van Wilgen and Biggs, 2011; Biggs and Potgieter, 1999). In some cases, low numbers, small areas and urgent time frames place severe constraints on experimental design, so that a succession of ‘trial and error’ evaluations, titrating progressively more interventionist management actions, offer the only practical insights for adaptation of management strategies (Briceño-Lenares et al., 2011).

2.4. Institutional barriers

Adaptive management programs typically take place in settings where multiple institutions (“stakeholders”) have at least some management responsibility and authority. Further, such institutions offer severe barriers to development and implementation of adaptive policies, through factors ranging from differences in goals (and agency mandates) to inappropriate incentive systems (e.g. there is strong incentive in some government agencies to avoid taking leadership responsibility for building an adaptive management program). Such problems typically increase with institution size and system complexity, although these are arguably the circumstances where adaptive management approaches are most needed (Duncan and Wintle, 2008). Kingsford et al. (2011) give examples of institutional fragmentation where responsibilities for water management and supply are segregated, creating a need for leadership to develop strategies for meeting the contrasting objectives in a sustainable manner.

2.5. Models for policy screening

Models are imprecise representations of complex ecological systems, and at best can typically only predict directions of change in response to management policies. Yet models can be extremely useful tools that help clarify management problems and identify important knowledge gaps that question faith in model predictions. An initial modeling step can also help to “screen” policies that are simply too weak, too isolated in effect, or too costly to be considered candidates for further adaptive management experimentation (Walters, 1997). Models used this way are far less suspect than those intended to make precise quantitative predictions (which typically fails). Probert et al. (2011) use simple models to explore candidate management policies for which the objectives span a spectrum from maximising system benefit to maximising pure learning. By valuing knowledge gain and system benefit independently, they demonstrate a framework that allows the strengths of alternative policies to be compared, given the relative importance of learning compared to system benefit in the management objective.

2.6. Innovative monitoring approaches

Adaptive management, especially when it involves large scale and replicated experimental policy treatments, typically results in very high or unacceptable monitoring costs over long time periods. A critical need is to develop less costly (per observation) monitoring schemes using approaches such as stakeholder involvement in data gathering. Briceño-Lenares et al. (2011) show how monitor-

ing of both ecological and social system components can be achieved under severe logistic and economic constraints. Surrogate measures of biotic response have also been proposed and implemented to reduce monitoring costs (e.g. pyrodiversity as a surrogate for biodiversity in fire management, see van Wilgen and Biggs, 2011). Unless the efficacy of surrogates is tested against a sample of primary responses, however, there can be a substantial risk that monitoring will mis-inform management (Parr and Andersen, 2006). Perhaps the most common cause of inflated costs in monitoring programs is that resources are committed to monitoring things that are irrelevant or insensitive to system response, at levels of precision that are unnecessary. An *a priori* value of information analysis can help identify the important factors to monitor, as well as the level of precision that might be warranted (Runge et al., 2011; McDonald-Madden et al., 2010).

2.7. Conflict resolution

Most ecological management situations involve multiple, conflicting stakeholders and objectives. Debate about appropriate objectives and negative impacts on particular stakeholders can lead to paralysis in adaptive management programs, with implementation of policies that represent tolerable compromises that serve no one well. Kingsford et al. (2011) note that management becomes dysfunctional when disaffected stakeholders attempt to discredit the process or broaden objectives to an unworkable degree. Resolution of such conflicts can open doors to better solutions for all the other problem areas mentioned above. A variety of approaches are available to assist resolution including multi-attribute utility functions (Probert et al. (2011), Conroy et al., 2011); hierarchies of objectives (Kingsford et al., 2011) and hierarchical views of system dynamics and management (Conroy et al., 2011); economic incentives for cooperation (Briceño-Lenares et al., 2011); value of information assessment (Runge et al., 2011); and ongoing negotiation among stakeholders (van Wilgen and Biggs, 2011). Colyvan et al. (2011) argue that consensus is often an inappropriate model for resolving management conflicts. They evaluate game theory as a tool for understanding the interactions between parties with different interests and objectives. This highlights the phenomenon where policy choices most often represent not the objectives of a single decision maker, but rather the outcome of a gaming process among stakeholders with deeply conflicting objectives. Game theory helps to determine what decision strategies optimize conservation outcomes, irrespective of whether the parties are ‘playing’ to co-operate or not. It does so by providing insight into which strategies are likely to be employed by different interest groups, the types of co-operation that best reflect their respective objectives, and the range of possible outcomes under cooperative and non-cooperative involvement (Colyvan et al., 2011).

2.8. Self-serving behaviour

Another key barrier to implementation is self-serving behaviour by scientists and managers. Scientists often oversell their ability to model and predict policy consequences, and sometimes use policy demands to pursue discovery goals that may never be applied to decision making by testing their predictions. Managers often willfully ignore scientific uncertainty to avoid complicating their message to stakeholders, and use it (or the pretence of certainty) later when policies fail in attempts to shift responsibility for failed policy implementation to the scientific community. Both self-serving pursuits lead to misunderstanding about limits to science. The non-cooperative strategies that different stakeholders might employ, possible interactions between them, and how conservation decisions are structured have a significant influence on what con-

ervation goals are achievable (Colyvan et al., 2011). There is a critical need for scientists to shift thinking from discovery alone to decision making (Conroy et al., 2011) and for policy-makers to embrace uncertainty in assessing the relative risks and robustness of alternative policies as decisions are made (Walters, 2007).

3. Conclusion

Some papers in this issue present novel sophisticated approaches and tools that will support the model development and decision-making components of adaptive management. When implemented, these will help to make management decisions more robust to uncertainty; research developments, however, continue to outstrip practical applications of adaptive management (Allen and Stankey, 2009). This is most evident in complex ecosystems and institutional environments where adaptive approaches are most needed. Kruger National Park is one terrestrial landscape where the approach is well established and formally adopted at an institutional level (van Wilgen and Biggs, 2011), yet even in Kruger there are deficiencies in implementation that remain to be resolved. Here and elsewhere, a key hurdle is the need to embrace uncertainty, so that management effort is spread over competing options in order to learn about them experimentally, rather than over-investing in one option that is currently perceived as 'best practice'. In addition to other barriers discussed above, Duncan and Wintle (2008) identified stakeholder risk aversion, failure to identify clear and measurable management goals, inappropriate statistical approaches to inference and learning, failure to monitor the performance of alternative management actions and reluctance to invest in long-term management experiments as impediments to successful adaptive management. The challenge to researchers is to shift their focus from discovery to the science of implementation, while managers and policy-makers must depart from their socio-political norms and institutional frameworks to embrace new thinking and effectively utilise the wealth of powerful new scientific tools for learning by doing.

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